Module - 2

The ML workflow & habits.

⇒ [How to frame a Problem.]

→ Features & labels

→ Training | validation | Test [very very concept]

→ Baselines.

Theory in practice

→ Iris data :- (Setosa, versicolor, virginica) → classification

(P1, PW, SL, SW) → Data.

(1000 records)

extracting patterns

→ Titanic dataset. EDA based on all columns data

↳ [Survived or not.]

→ you you are a employee in Car24. (website)

(it give you a price)

(Domain expertise) (Build ?)

(talk with some who has there expertise)

[What kind of data i need to have?]

→ car release year

→ Brand

→ model

→ km

→ any

→ ederly

→ type

→ insurance

→ previous accidents.

estimate a Price

(collect the data from database) (hardest Part).

	car_name	brand	model	vehicle_age	km_driven	seller_type	fuel_type	transmission_type	mileage	engine	max_power	seats	selling_price
0	Maruti Alto	Maruti	Alto	9	120000	Individual	Petrol	Manual	19.7	796	46.3	5	120000
1	Hyundai Grand	Hyundai	Grand	5	20000	Individual	Petrol	Manual	18.9	1197	82	5	550000
2	Hyundai i20	Hyundai	i20	11	60000	Individual	Petrol	Manual	17	1197	80	5	215000
3	Maruti Alto	Maruti	Alto	9	37000	Individual	Petrol	Manual	20.92	998	67.1	5	226000
4	Ford Ecosport	Ford	Ecosport	6	30000	Dealer	Diesel	Manual	22.77	1498	98.59	5	570000
5	Maruti Wagon R	Maruti	Wagon R	8	35000	Individual	Petrol	Manual	18.9	998	67.1	5	350000
6	Hyundai i10	Hyundai	i10	8	40000	Dealer	Petrol	Manual	20.36	1197	78.9	5	315000
7	Maruti Wagon R	Maruti	Wagon R	3	17512	Dealer	Petrol	Manual	20.51	998	67.04	5	410000
8	Hyundai Venue	Hyundai	Venue	2	20000	Individual	Petrol	Automatic	18.15	998	118.35	5	1050000
12	Maruti Swift	Maruti	Swift	4	28321	Dealer	Petrol	Manual	16.6	1197	85	5	511000
14	Hyundai Verna	Hyundai	Verna	8	65278	Dealer	Diesel	Manual	22.32	1582	126.32	5	425000
15	Renault Duster	Renault	Duster	5	50000	Individual	Diesel	Manual	19.64	1461	108.45	5	750000
16	Mini Cooper	Mini	Cooper	4	6000	Dealer	Petrol	Automatic	14.41	1998	189.08	5	3250000
17	Maruti Ciaz	Maruti	Ciaz	5	78000	Dealer	Diesel	Manual	28.09	1248	88.5	5	650000

→ ① what are you predicting? (selling price)

↳ Target column

↳ dependent variable. feature

remaining cols → independent features.

→ There is a lot of variation in data types.

→ is the target a Number or a category

↳ Regression Problem

Problem

Classification

Regression Problem.

[binary classification]

[Yes or No
0 or 1]

↳ Titanic data.

categorical classification

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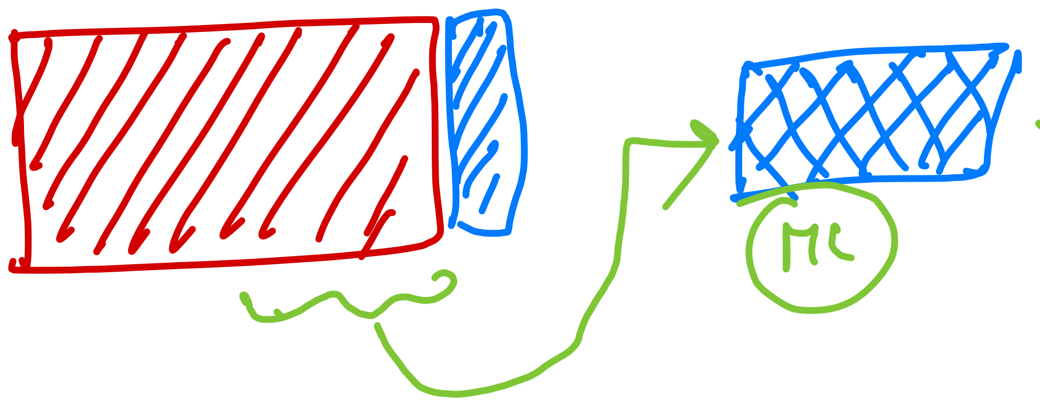
Nomenclature

X ?

inputs
independent

y

output
dependent



ML

(Trained model).

Predict selling price

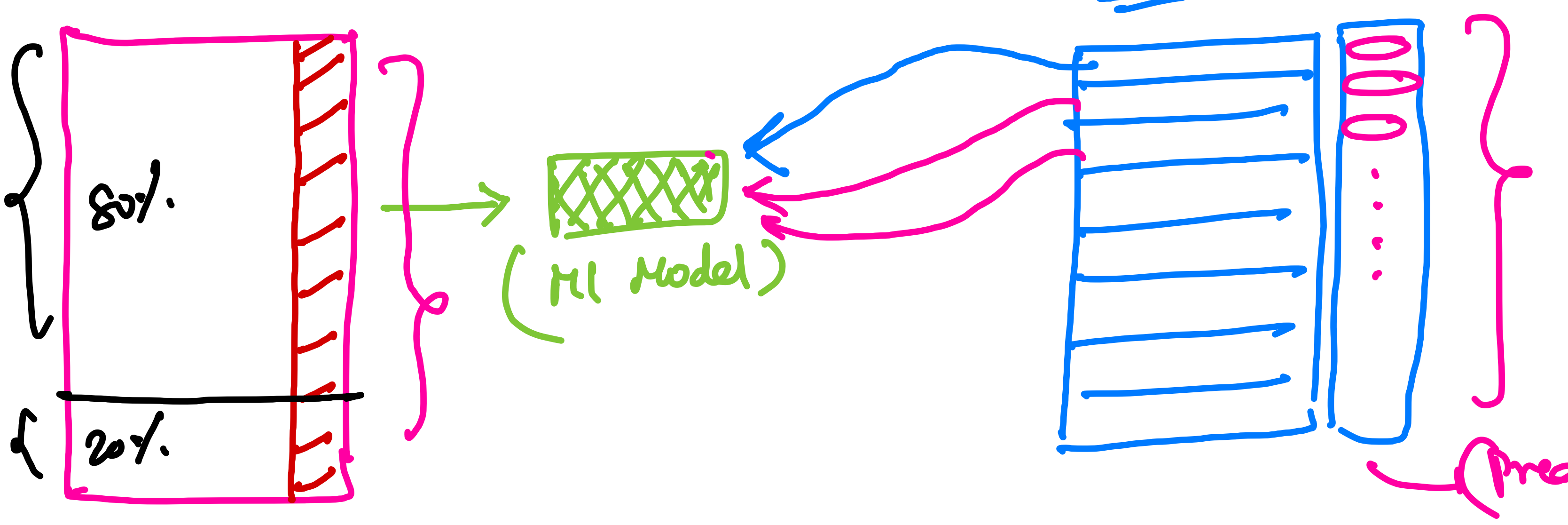
500,000

645,000

15000 + records dataset

Future

don't have target column



80%

20%

15000.

Past data

(Ml Model)

good or bad ?

(Predictions)

Split the data (training)

Test train split 70% 30%

60 : 40%

12000 80% Training → ML model → SP

Calculate Average error MAE

3000 20% Testing

I will keep it seperately.

Test	Pred	diff
10	8	2
12	5	7
14	12	2
8	10	2

⇒ mean = 3.25

→ Data leakage:

① → College exam → 95% → Impressive

accidentally gave you the answer key → 95%

(Data leakage)

did you learn anything?

Testing → model will provide very good result

Training

80% Train

20% predict

Testing data

model evaluation

good procedure

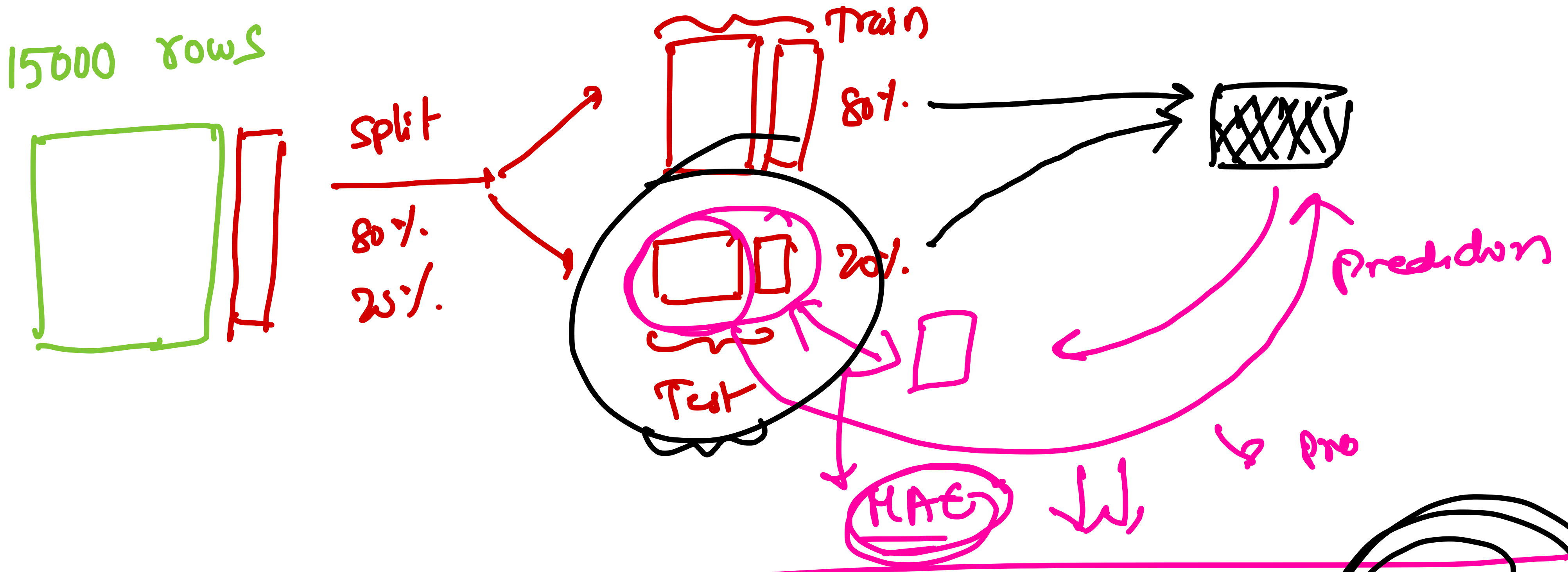
knows this data

evaluation

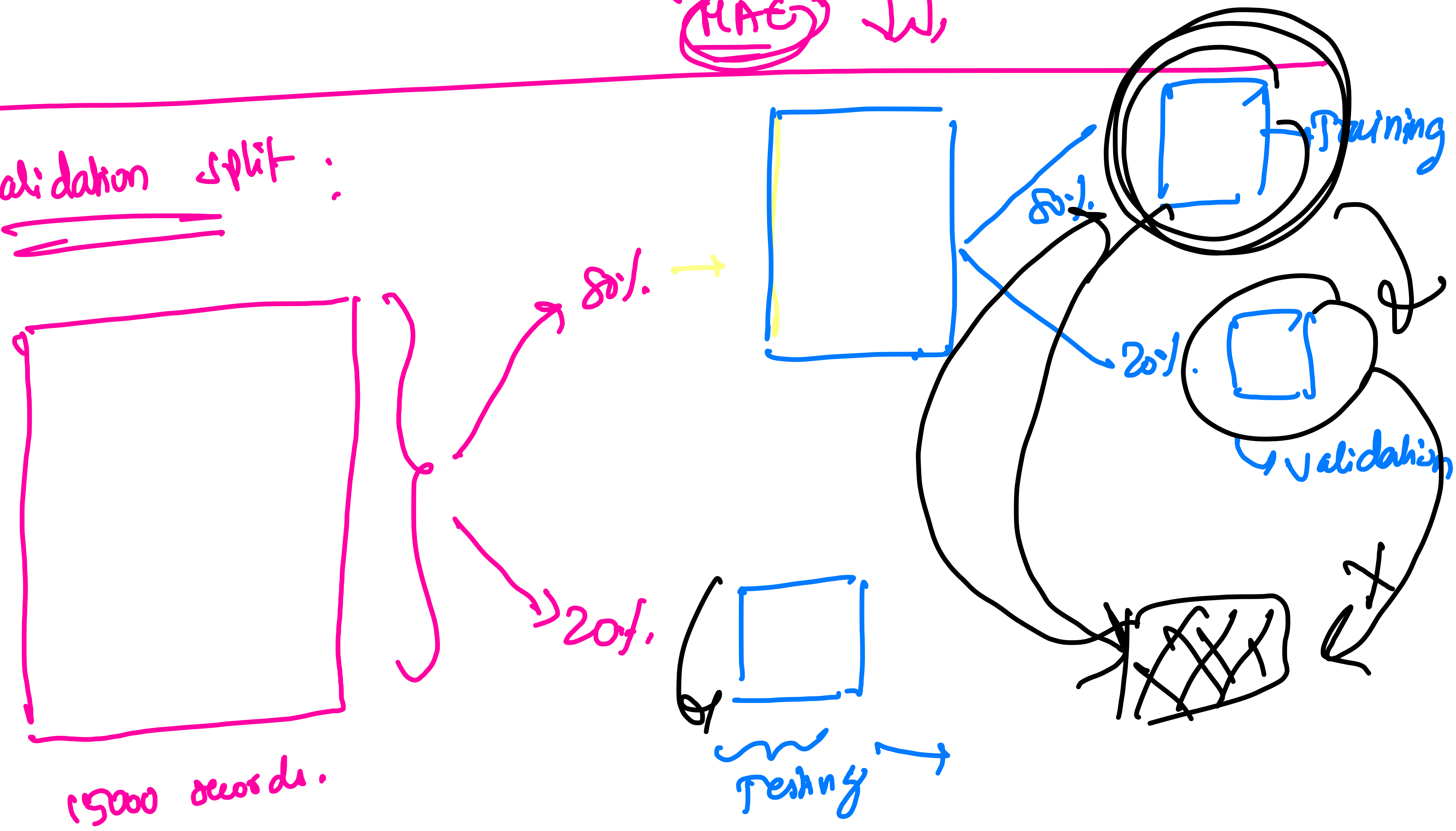
predictions

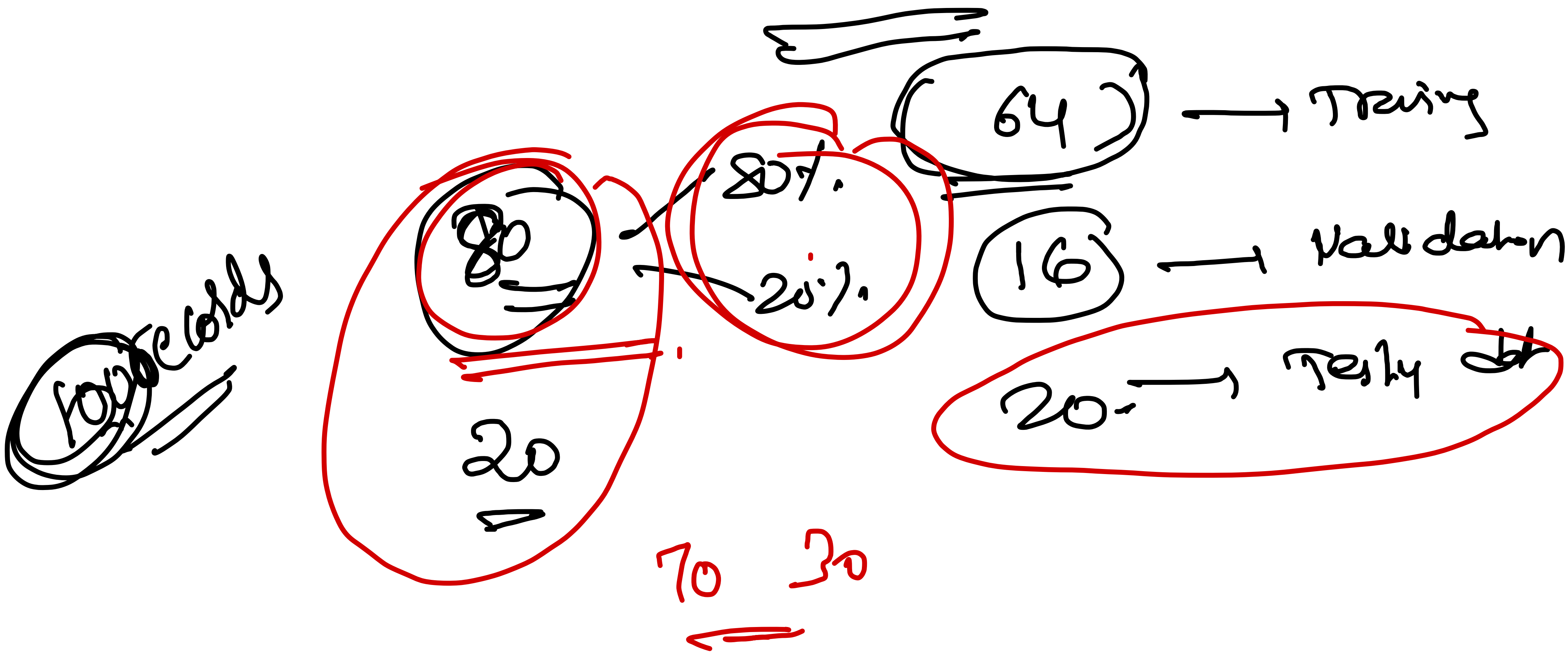
MAE =

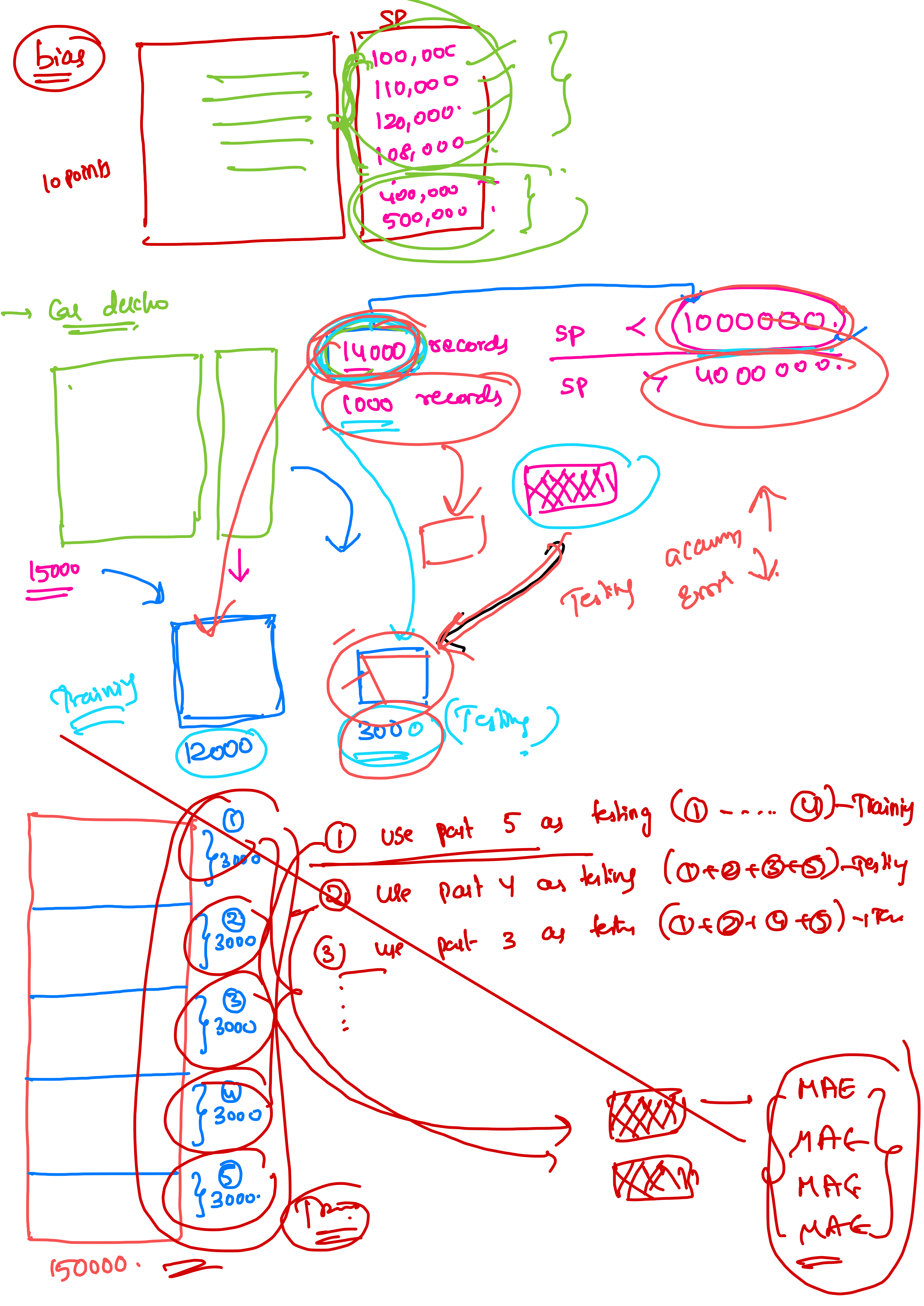
2000 rupees



validation split :






bias
10 points
SP
100,000
110,000
120,000
108,000
400,000
500,000
14000 records
1000 records
SP < 1000000.
SP > 4000000.
15000
Training
12000
3000
(Testing)
Testing
accuracy
Error
① use part 5 as testing (① .... ④)—Training
② use part 4 as testing (①+②+③+⑤)—Testing
③ use part 3 as testing (①+②+④+⑤)—Tra
① 3000
② 3000
③ 3000
④ 3000
⑤ 3000.
150000.
MAE
MAE
MAE
MAE

Summary

Concept	What It Is	Key Takeaway
Machine Learning	Finding patterns in old data to predict new data	Not magic — just pattern-finding at scale
Problem Framing	Defining what you predict, with what, and for whom	Get this right first — everything else depends on it
Features (X)	The input columns used to make predictions	Everything EXCEPT the target. Drop IDs and the target itself
Label/Target (y)	The column you're trying to predict	`selling_price` in our case
Regression	Predicting a continuous number	Our problem — price is a number
Classification	Predicting a category	Not our problem (yet) — coming later
Train/Test Split	Separate data into learning set and evaluation set	Split BEFORE any preprocessing. Test set is sacred — touch once
Baseline	The simplest possible prediction (mean/median)	Every model must beat the baseline to be useful
MAE	Mean Absolute Error — average prediction error in ₹	"On average, predictions are off by ₹___"

→ MAE :- mean absolute error.

predictions

Abs |(y − y pred)|

1
2
3
4
5

Target	predictions	Abs
150,000	145,000	+ 5000
300,000	290,000	+ 10,000
4,50,000	450,000	− 0
1,70,000	150,000	+ 20,000
2,90,000	3,30,000	− 40,000

Input (X)

Target (Y).

Collected data.

10,000 records

100,000 record.

(−40,000)

Y − (Y_Pred)

Average : 15000

−5000/5 = −1000

total = 75000 / 5 = ₹15000

MAE = sum(y − ypred)/total records)

2

abs(2) = 2

abs(−2) = 2

10,000 → MAE (2)

$$\boxed{MAE = sum(|y - ypred|) / total\_num\_records.}$$

→ **Baseline** :-

input	y	$\hat{y}$
1	150,000	145,000
2	300,000	290,000
3	4,50,000	450,000
4	1,70,000	150,000
5	2,90,000	3,30,000

MAE = 15000

Avg = $\boxed{272,000}$

accuracy 60%

$\hat{y}_1$	$\lvert y - \hat{y}_1 \rvert$
272,000	+122000
272,000	28,000
272,000	178,000
272,000	+102,000
272,000.	18,000

average = 468,000 / 5

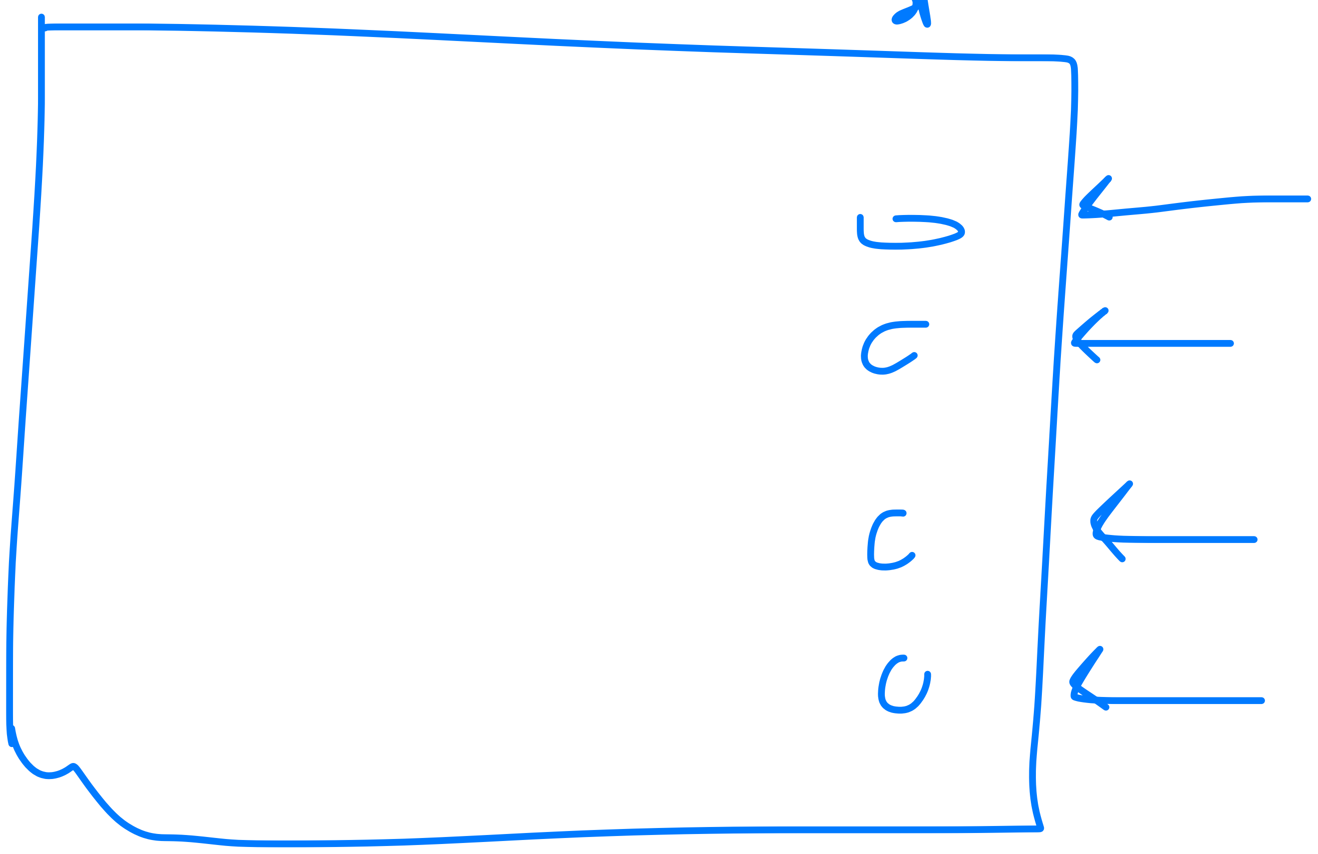
$\boxed{\text{MAE} \Rightarrow 93,600}$

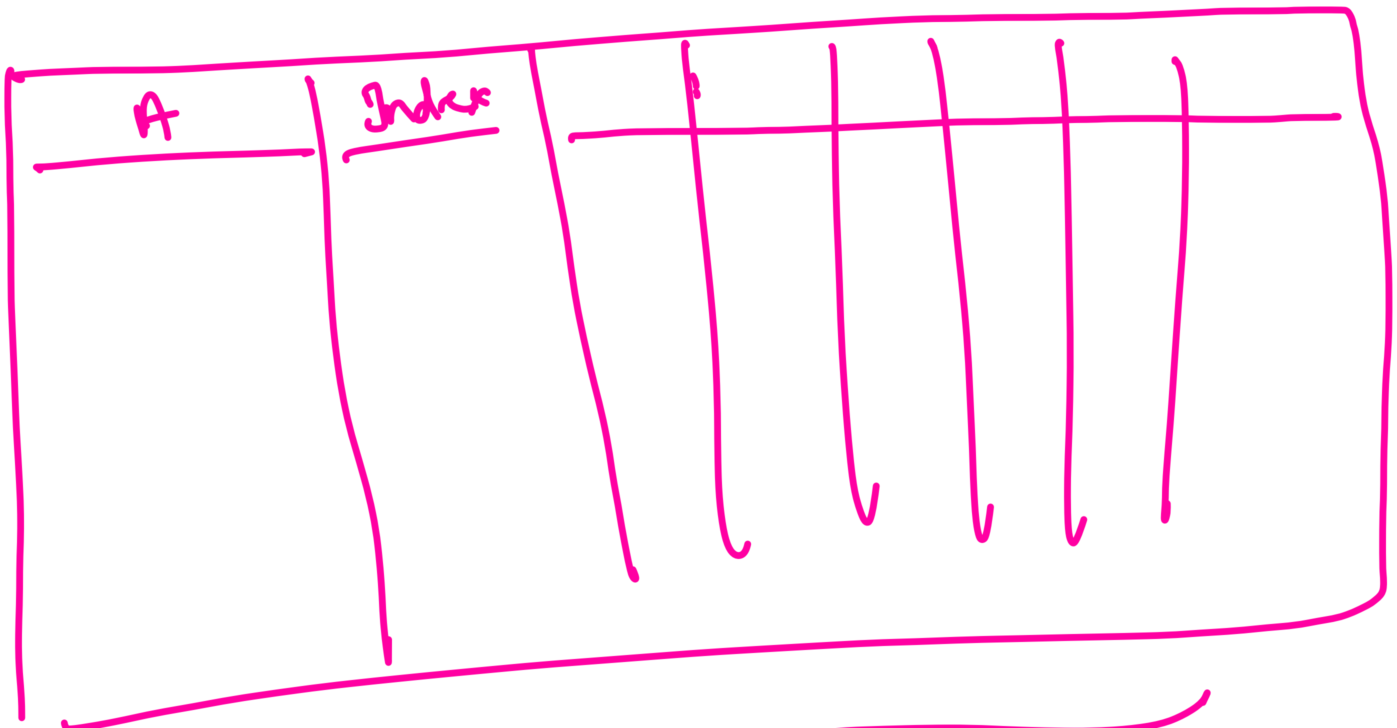
Baseline

80,00

79,00

SP





SQL table

Wher condition =

" . . . . 00314 }

via index

How much data is missing?

<5%.

(Drop the records)

5-30%.

(Imputation)

(Impute)

(Smart)

>40%.

Drop that column all together)

5 lakhs

11 lakhs

car

max variant

(mid variant / base)

999CC

(median)

(lets try to impute all the columns with median value)

seats → mode (→ 5 seats | 7 seats)

→ Before we start imputing.

MCAR → missing completely at Random | mean/median.

MAR → Missing at Random

MNAR → Missing not at Random.

(Hardest to fill)

year

2000

Sunroof

Yes/No

Encoding Techniques :-

↳ The process of converting English / Text into numericals.

→ Label encoding → $\boxed{\text{manual} = 0, \text{automatic} = 1}$ to each category.

↑ assigning a number

One-Hot encoding

Target / Frequency Encoding.

One hot encoding :-

Petrol, Diesel, CNG, LPG, electric. → 5 unique values.

fuel_Diesel

(N-1) column

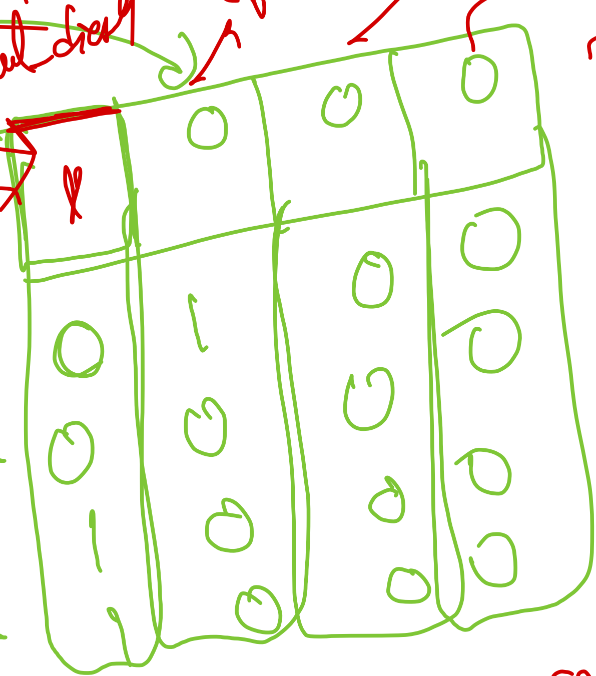
all of them are unique

	fuel_diesel	fuel_cng	fuel_lpg	fuel_elect
Petrol	0	0	0	0
Diesel	1	0	0	0
CNG	0	1	0	0
LPG	0	0	1	0
electric	0	0	0	1

brand	model	vehicle_age	km_driven	seller_type	fuel_type	transmission_type
Renault	Duster	7	67647	Dealer	Diesel	0
Maruti	Wagon R	3	52000	Dealer	Cng	0
Honda	City	3	48959	Dealer	Petrol	0
Mahindra	Scorpio	6	82000	Dealer	Diesel	0
Mahindra	XUV500	4	34000	Dealer	Diesel	1

1

fuel_diesel fuel_cng fuel_lpg fuel_elect



fuel_Petrol = 0, 0, 0, 0

cng = 0, 0, 0, 0

e	~~fuel_type_Diesel~~	~~fuel_type_Electric~~	~~fuel_type_Lpg~~	~~fuel_type_Petrol~~
0	1 True	0 False	0 False	0 False
0	0 False	0 False	0 False	0 False
0	0 False	0 False	0 False	1 True
0	1 True	0 False	1 False	0 False
0	True	False	False	False
..	...	...	...	...
0	True	False	False	False
0	False	False	False	True
0	False	False	False	True
0	False	False	False	True
0	False	False	False	True

animal

Dog, Cat, Rabbit → numericals.

(n-1) =

0, 1

| 0 | 0 | 0 |

| 1 | 0 | 0 |

| 0 | 1 | 0 |

| 0 | 0 | 1 | → ?

Dog

| 0 | 0 |

animal Cat, animal rabbit

| 1 | 0 |

Rabbit

| 0 | 1 |

| 1 | 1 | ?

age	income
10	—
2	1800
3	2000
9	—
12	999 CC
	⋮

9.5